

Shaurya Tiwari

AI Software Engineer • Physical AI • OpenUSD • Simulation

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SUMMARY

AI Software Engineer shipping production agentic systems and translating cross-functional requirements into deployed solutions. NVIDIA-Certified in OpenUSD with hands-on Isaac Sim, Isaac Lab, robot policy training, and simulation workflows for Physical AI.

PHYSICAL AI & SIMULATION

NVIDIA-Certified Professional: OpenUSD Development
Verify on Credly

July 2026

Force-Aware Peg Insertion

Isaac Lab, Isaac Sim, PPO, PhysX [GitHub ↗](#)

- Benchmarked four PPO policy variants across three seeds to evaluate whether force feedback improves Franka peg-insertion robustness under nominal and randomized conditions.
- Designed controlled reward/observation ablations with TensorBoard-logged metrics against a reproduced geometry-only baseline.

GPU-Parallel Robot Policy Training

Isaac Lab, Isaac Sim, PhysX, ovphysx, Python [GitHub ↗](#)

- Built GPU-parallel reinforcement-learning workflows in Isaac Lab, running large-scale PhysX simulations for robot policy training.
- Configured vectorized environments, observations, actions, rewards, and domain randomization for high-throughput policy training and evaluation.

Digital-Twin & Robotics Asset Pipeline

Omniverse, OpenUSD, ovrnx, ovphysx, Isaac Sim [GitHub ↗](#)

- Assembled and optimized SimReady OpenUSD digital twins and robotics assets using composition, instancing, materials, URDF interoperability, validation, and scene-optimization workflows.
- Structured USD environments for modular *ovrnx* rendering/sensor simulation and *ovphysx* physics workflows across NVIDIA Physical AI applications.

EXPERIENCE

AI Software Engineer

Aug 2024 - Present

U.S. News & World Report, New York, NY

- Partnered with stakeholders across 12 business verticals to convert ambiguous domain problems into validated AI roadmaps, implementation plans, and production-ready engineering scopes.
- Architected and maintained ByteSage, a Django backend powering LangGraph-based agentic workflows with multi-tool orchestration, conditional routing, streaming, and production guardrails.

Software Engineer

May 2023 - May 2024

COAPS, Florida State University, Tallahassee, FL

- Engineered scalable ETL pipelines processing 1M+ marine data points into PostgreSQL-backed systems supporting large research datasets.

TECHNICAL PROJECTS	AutoTicket March 2026 <i>Internal Company Hackathon</i> Built an automatic Jira/Confluence/GitLab-integrated triaging system on ByteSage's agent orchestration, generating tickets and boilerplate code from intake forms and saving up to 3 weeks of project initiation time.
	POView March 2026 <i>Cerebral Valley Hackathon</i> Built a 3D geospatial globe interface with the Google Places API and Gemini Live API, letting users navigate to referenced locations through conversational AI.
	InstruX Feb 2026 - Present <i>Open Source Project</i> instrux.world ↗ Designed a reference architecture for an agentic robotics-evaluation layer connecting simulation, synthetic scenario generation, and policy-readiness assessment.
NVIDIA APPLIED WORK	Technical Communication 100+ livestreams hosted on OpenUSD, Isaac Sim, Isaac Lab, and Physical AI concepts. All logged on YouTube Live with code, notes, and references.
	SIGGRAPH 2026 — NVIDIA Physical AI Hands-On Labs Blender, Oranges, and ovrtnx: Integrated ovrtnx, ovphysx, and ovstage with Blender to build an RTX-rendered, physics-enabled OpenUSD workflow using composable NVIDIA Omniverse Libraries. Ray Trace Your Way With Omniverse Libraries: Built an OpenUSD-based RTX viewport with Omniverse Libraries, loading and querying USD scenes, rendering RTX output, controlling cameras, and generating image results. Advancing World Simulation With 3D Gaussian Splatting: Built a real-world-to-simulation reconstruction workflow using NVIDIA fVDB and Omniverse NuRec, applying multi-GPU 3D Gaussian Splatting, segmentation, and object-level integration to create simulation-ready environments for Isaac Sim / Isaac Lab.
ADDITIONAL EXPERIENCE	Graduate Research Assistant Aug 2022 - Apr 2023 <i>Department of Scientific Computing, Florida State University, Tallahassee, FL</i> Built and trained deep learning models in PyTorch, including GAN-based experiments and reinforcement learning concepts for synthetic climate-state data generation.
	Junior DevOps Engineer Jan 2021 - Dec 2021 <i>ACS Pvt. Ltd., Mumbai, India</i> Maintained CI/CD pipelines with Jenkins and Docker, reducing MTTR by 20% through CloudWatch-based monitoring and alerts.
SKILLS	<i>Physical AI / Simulation:</i> OpenUSD, Isaac Sim, Isaac Lab, PhysX, ovrtnx, ovphysx, ovstage <i>Programming:</i> Python, C, JavaScript/TypeScript, SQL <i>AI / ML:</i> PyTorch, LangGraph, LangChain, PPO, RAG <i>Backend / Cloud:</i> Django, AWS, Docker, PostgreSQL, OpenSearch
EDUCATION	M.S. in Computer Science, GPA 3.8/4.0 May 2024 <i>Florida State University, Tallahassee, FL</i>
	B.Tech in Information Technology, GPA 3.3/4.0 Aug 2022 <i>NMIMS University, Mumbai, India</i>